

Georgiy Lavrentev

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EDUCATION

ITMO University — Applied Mathematics and Computer Science 2024 – present
Software engineering, algorithms, systems programming.

EXPERIENCE

Open Position — Backend / DevOps seeking
A position at your company would be my first professional work or internship experience.

PROJECTS

Social network model Go, gRPC, Protobuf, PostgreSQL, HTML, Docker
Model of a social network: Go backend with gRPC/Protobuf, HTML frontend, PostgreSQL in Docker.

Concrypto (Concurrency cryptography) Go, AES-GCM, PCI-DSS, crypto/aes, crypto/cipher, sync
Multithreaded encryption of bank cards using AES-GCM with PCI-DSS key rotation.

FAT-manager C/C++, Qt, QMake, clang-format
FAT16/FAT32 filesystem parser with a dual-panel UI inspired by Far Manager.

Telegram bot for the social network Go, Python, Telegram Bot API
Bot interface for the social network model: command handling, Telegram API integration.

TECHNICAL SKILLS

Languages: Go, Java, Python, C/C++
Frameworks: gRPC, Protobuf, Telegram Bot API, Matplotlib, Qt
Developer Tools: Docker, Git, Linux, QMake, clang-format

Social network model

Stack: Go, gRPC, Protobuf, PostgreSQL, HTML, Docker

About

Building a social network model. Initially a Telegram bot served as the UI (users logged in via tg_id); later a web interface was added. Backend logic is written in Go using Protobuf code generation and gRPC for inter-service communication. Data is stored in PostgreSQL running in a Docker container.

What was done

- Two APIs: one for the Telegram bot, one for the web interface
- gRPC services with Protobuf code generation
- PostgreSQL in Docker: SQL queries, migrations
- Shell scripts for launching the full app and individual components

Challenges

The main difficulty was designing the API for the web interface. Running the backend on a remote server without a VPN caused requests to be very slow or unresponsive. The chosen solution was to use a VPN, which is acceptable given the project's educational scope.

Concrypto (Concurrency cryptography)

Stack: Go, AES-GCM, PCI-DSS, crypto/aes, crypto/cipher, sync

Background

A bank-themed task: card data must not be stored or transmitted in plaintext. PCI-DSS requires periodic key rotation to limit the impact of a key compromise. The goal was to implement a concurrent card-encryption function with key rotation support.

What was done

- Card encryption using AES-GCM algorithm
- Key rotation logic in the card vault
- Concurrent implementation using the sync package
- Libraries: crypto/aes, crypto/cipher, crypto/rand, encoding/hex

Note

Repository is shared privately on request — redistribution is prohibited by course rules.

FAT-manager

Stack: C/C++, Qt, QMake, clang-format

About

A parser for FAT16 and FAT32 filesystem images, presented as a two-panel file manager inspired by Far Manager: local storage on the left, the FAT image on the right.

What was done

- FAT16 and FAT32 support
- Two-panel Qt UI
- QMake project setup
- Code formatting with clang-format

Note

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Telegram bot for the social network

Stack: Go, Python, Telegram Bot API

About

A Telegram bot as the primary interface for the social network model. Users authenticate via their Telegram ID and interact with the system through bot commands. Integrated with the Go backend.

What was done

- Command and message handling
- Authentication via Telegram ID
- Telegram Bot API integration
- Communication with the social network backend services